Customer Shopping Behaviour Analysis

Transactional analysis of **3,900 purchases** across product categories — uncovering patterns in customer segments, product preferences, and subscription behaviour.

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Dataset at a Glance

3,900	18	$59.76	3.75
Purchases	Columns	Avg. Purchase	Avg. Rating

Demographics

Age, Gender, Location, Subscription

Purchase Details

Item, Category, Amount, Season, Payment

Behaviour

Discounts, Frequency, Reviews, Shipping

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Data Cleaning in Python

01

Load & Explore

Imported via `pandas`; inspected with `df.head()`, `df.info()`, `df.describe()`

02

Handle Missing Data

37 null values in *Review Rating* — imputed with category median

03

Feature Engineering

Added `age_group` bins and `purchase_frequency_days` numeric column

04

Clean & Export

Removed redundant `promo_code_used`; loaded to PostgreSQL

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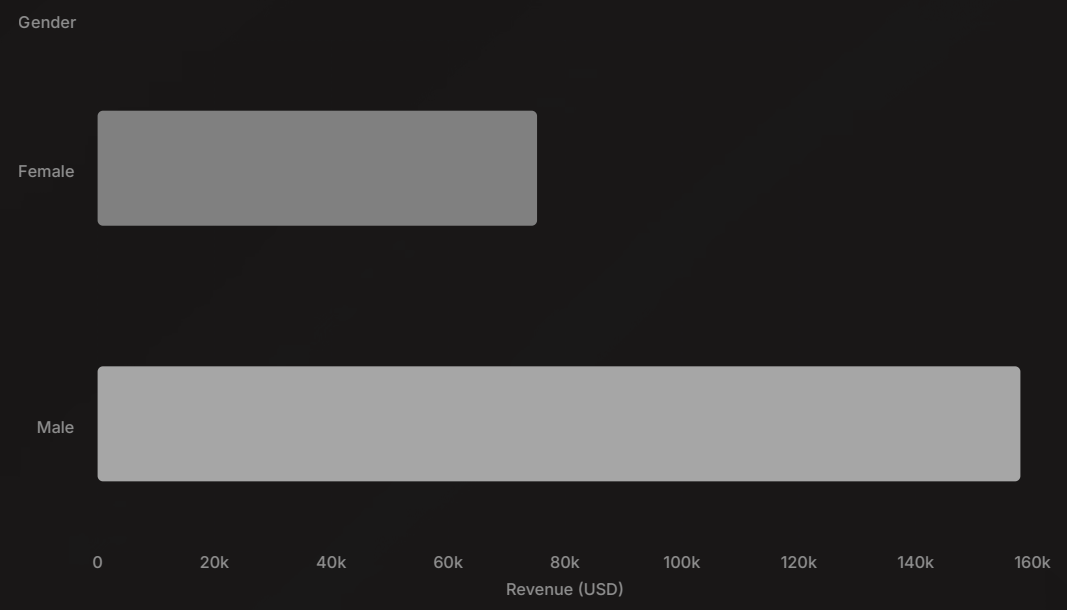
SQL Analysis: Revenue & Spending



Key Findings

Male customers generated **2× the revenue** of female customers.

839 discount users still spent above the $59.76 average — discounts don't always suppress spend.

Express shipping users averaged **$60.48** vs. $58.46 for Standard.

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Top-Rated Products & Discount Dependence

Top 5 by Avg. Rating

Product	Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent

Product	% Discounted
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

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Customer Segmentation & Loyalty

3,116	701	83
Loyal	**Returning**	**New**
>10 previous purchases	2–10 previous purchases	1 previous purchase

80% of the customer base is Loyal — a strong foundation for retention-focused strategy.

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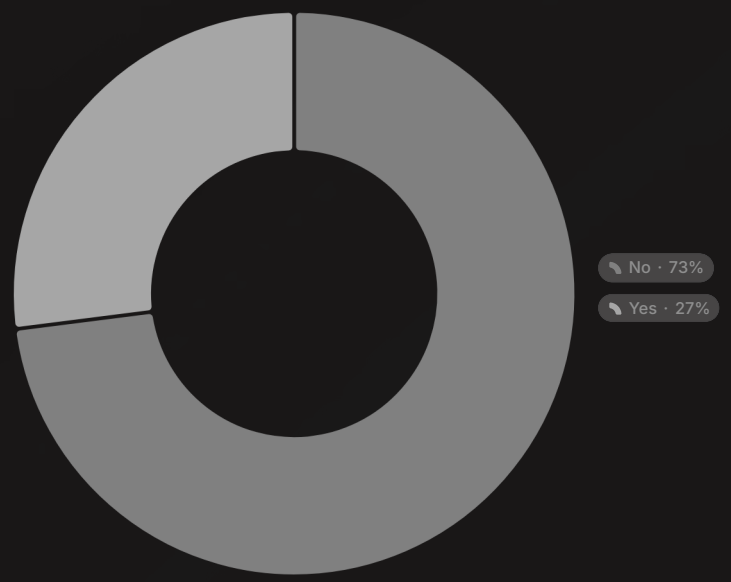
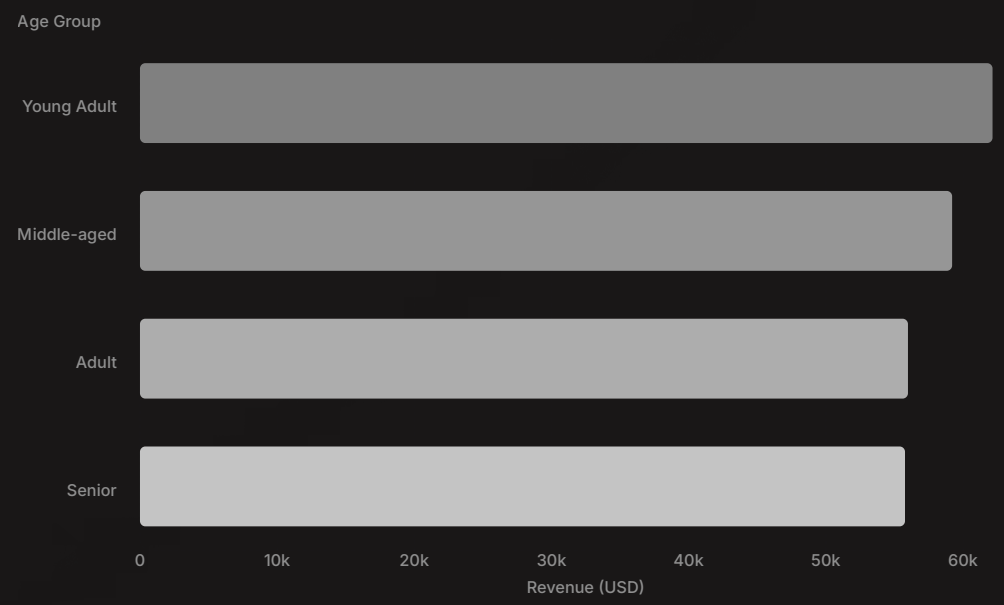
Subscriptions & Revenue by Age





Only **27%** of customers subscribe. Avg. purchase is nearly identical: subscribers $59.49 vs. non-subscribers $59.87.

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Power BI Dashboard

Interactive Power BI dashboard visualising subscription status, revenue by category, sales by age group, and key KPIs.

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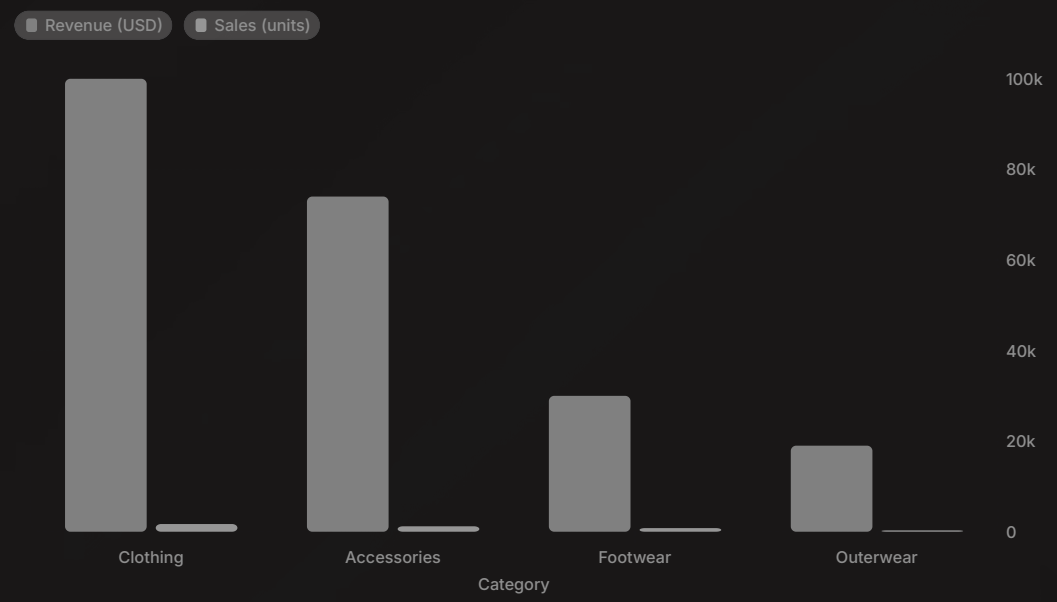
Revenue by Category



Top Products Per Category

- **Accessories:** Jewelry, Sunglasses, Belt
- **Clothing:** Blouse, Pants, Shirt
- **Footwear:** Sandals, Shoes, Sneakers
- **Outerwear:** Jacket, Coat

Clothing dominates both revenue and volume.

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the **73%** non-subscribers

Loyalty Programs

Reward repeat buyers to drive subscription uptake

Review Discount Policy

Balance revenue uplift with margin control

Product Positioning

Highlight top-rated and best-selling items in campaigns

Targeted Marketing

Focus on Young Adults and Express-shipping users

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